ALGEBRA 2 · TOPIC 3 · LESSONS 3-1 AND 3-2

Name _____

Topic 3 Review: Lessons 3-1 and 3-2

Date _____ Period _____

Show your work. Answers are in the answer key that comes with this worksheet in Classboard.

A · CLASSIFY

1. Write in standard form, then give the degree and leading coefficient: $9 - x^2 + 4x^3$

2. Name by degree and number of terms: $-6x^4 + x$

3. Is $3x^2 + 2/x$ a polynomial? Explain.

4. Write a cubic trinomial with a leading coefficient of −2.

B · END BEHAVIOR AND TURNING POINTS

5. Describe the end behavior of $f(x) = -3x^4 + 2x - 1$.

6. Describe the end behavior of $g(x) = 0.5x^5 - x^3$.

7. What is the most turning points $h(x) = x^6 - 2x^2 + 1$ can have? The most real zeros?

8. A polynomial graph rises on the left, falls on the right, and has 2 turning points. Give a possible degree and the sign of its leading coefficient.

C · ZEROS AND GRAPHS

9. For $f(x) = (x + 4)(x - 1)(x - 3)$: list the zeros, the y-intercept and the end behavior.

10. Find the average rate of change of $f(x) = x^3 - 2x$ from $x = -1$ to $x = 2$.

11. For $p(x) = x^3 - 9x$: factor it, find the zeros, then use $p(-2)$ and $p(2)$ to say where the relative max and min are.

D · ADD AND SUBTRACT

12. $(2x^3 + 5x^2 - x) + (x^3 - 5x^2 + 7)$

13. $(6x^2 - 4x + 3) - (2x^2 + 4x - 5)$

14. $(x^4 - 3x^2 + 2) - (x^4 + x^3 - 3x^2)$

15. $(4a^2 - 3ab + b^2) - (a^2 + ab - 2b^2)$

E · MULTIPLY

16. $(3x + 2)(x - 4)$

17. $(x - 1)(x^2 + 3x - 2)$

18. $-2x(x+5)(x-1)$

19. $(x+3)^3$

F · APPLY IT

20. A box is made from a 14 cm by 9 cm sheet by cutting an x cm square from each corner. Write $V(x)$ in standard form and give the values of x that make sense.